Chapter 8

Exponents, Radicals & Exponential Functions

Imagine filling a chess board with grains of rice: one grain on the first square, two on the second, four on the third — each square doubling the last. By square 64, you would need more rice than has ever been grown on Earth. This astonishing growth is the power of exponents. In this chapter we will build the rules that govern that power, learn how to undo it with radicals, and explore the beautiful curves that exponential functions trace across the plane.

Section 8.1 The Laws (Properties) of Exponents

Section 8.2 Radical Notation & Rational Exponents

Section 8.3 The Exponential Function $y = 2^x$

Contents

1 Exponents, Radicals & Exponential Functions **5**
1.1 The Laws of Exponents 5
1.2 Radical Notation & Rational Exponents 7
1.3 The Exponential Function $y = 2^x$ 9
Chapter 8 Practice Worksheet 11
Answer Key 13

Chapter 1

Exponents, Radicals & Exponential Functions

1.1 The Laws of Exponents

Think of an exponent as a shortcut for repeated multiplication. Instead of writing $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$*, mathematicians invented the notation* 2^5*. Over centuries of working with these shortcuts, they discovered patterns — rules that always hold, no matter which numbers are involved. We call these the* Laws of Exponents.

Definition: Exponent and Base

Definition: Exponent & Base

For any real number a and positive integer n,

$$a^n = \underbrace{a \cdot a \cdot a \cdots a}_{n \text{ factors}}$$

Here a is the **base** and n is the **exponent** (or *power*). The exponent tells you *how many times* to multiply the base by itself.

The Five Core Laws

① PRODUCT RULE

$$a^m \cdot a^n = a^{m+n}$$

$$2^3 \cdot 2^4 = 2^7 = 128$$

② POWER-OF-A-POWER RULE

$$(a^m)^n = a^{m \cdot n}$$

$$(2^3)^4 = 2^{12} = 4096$$

③ ZERO EXPONENT RULE

$$a^0 = 1 \quad (a \neq 0)$$

$$2^0 = 1, \quad 99^0 = 1$$

④ NEGATIVE EXPONENT RULE

$$a^{-n} = \frac{1}{a^n}$$

$$2^{-1} = \tfrac{1}{2}, \quad 2^{-2} = \tfrac{1}{4}$$

⑤ QUOTIENT RULE

$$\frac{a^m}{a^n} = a^{m-n}$$

$$\frac{2^{10}}{2^3} = 2^7 = 128$$

Why does $a^0 = 1$***?*** *Use the Quotient Rule:* $\dfrac{a^n}{a^n} = a^{n-n} = a^0$. *But any nonzero number divided by itself equals* 1. *So* $a^0 = 1$. *The zero rule is not arbitrary — it follows directly from the Quotient Rule!*

Worked Example: 3.1.1 Simplify using the Laws of Exponents

Problem: Simplify $(2^3)^4 \cdot 2^{-5}$.

$$\begin{aligned}
(2^3)^4 \cdot 2^{-5} &= 2^{3\cdot 4} \cdot 2^{-5} && \text{(Power-of-a-Power Rule)} \\
&= 2^{12} \cdot 2^{-5} \\
&= 2^{12+(-5)} && \text{(Product Rule)} \\
&= 2^7 \\
&= \boxed{128}
\end{aligned}$$

Worked Example: 3.1.2 Negative Exponents

Write 3^{-4} as a fraction, then evaluate.

$$3^{-4} = \frac{1}{3^4} = \frac{1}{81}$$

1.2 Radical Notation & Rational Exponents

You may have noticed that the radical sign $\sqrt{\ }$ *looks exactly like the symbol for* long division. *That is no accident — it evolved from the letter "r" (for the Latin* radix, *meaning "root"). Radicals are the inverse of exponentiation, just as division is the inverse of multiplication. When we ask* $\sqrt{16}$*, we are asking:* what number, multiplied by itself, gives 16?

Definition: The n-th Root

Definition: n-th Root & Radical Notation

For a positive integer n and a real number a,

$$\sqrt[n]{a} = b \quad \Longleftrightarrow \quad b^n = a$$

- $\sqrt[n]{\ }$ is the **radical sign**
- n (written in the "crook" of the sign) is the **index**
- a (under the bar) is the **radicand**

When no index is written, it is understood to be 2 (the square root).

From Powers to Radicals

Worked Example: 3.2.1 Solving for x using a radical

Suppose we know that x multiplied by itself 10 times equals 1024. What is x?

$$\begin{aligned} x^{10} &= 1024 \\ x &= \sqrt[10]{1024} \quad \text{(take the 10th root of both sides)} \end{aligned}$$

We can check: $2^{10} = 1024$, so

$$\boxed{x = \sqrt[10]{1024} = 2}$$

In the expression above, 10 is the *index* and 1024 is the *radicand*.

Rational Exponents: The Bridge

Starting from $x^{10} = 1024$, we can raise both sides to the power $\frac{1}{10}$:

$$\left(x^{10}\right)^{1/10} = 1024^{1/10} \quad \Longrightarrow \quad x = 1024^{1/10}$$

But we already know $x = \sqrt[10]{1024}$. So the two expressions are identical:

Definition: Rational Exponent

$$a^{1/n} = \sqrt[n]{a}$$

More generally,

$$a^{m/n} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}$$

The **denominator** of the rational exponent becomes the *index* of the radical; the **numerator** becomes the *power*.

Worked Example: 3.2.2 Converting between forms

$8^{1/3} = \sqrt[3]{8} = 2$ (cube root of 8)

$\sqrt[4]{x^3} = x^{3/4}$ (index 4 becomes denominator, power 3 becomes numerator)

$27^{2/3} = \left(\sqrt[3]{27}\right)^2 = 3^2 = 9$ (find cube root first, then square)

1.3 The Exponential Function $y = 2^x$

Picture sitting in an airplane window seat. As the plane banks over a city, you look down and see the streets lined with trees — one long queue of green stretching to the horizon. Each tree is a point. If you could tag every point on that line with a number, you would have the number line, which is just the x-axis. Every one of those points x has a corresponding height $y = 2^x$. The set of all those heights, swept from left to right, is *the curve of the exponential function. All the straight lines and curves we draw in algebra are, in this sense, portraits of queues of points.*

Domain, Range, and Key Values

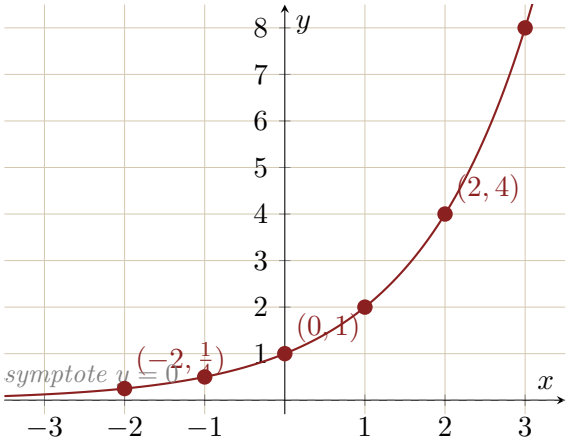
Definition: Key Facts about $y = 2^x$

- **Domain:** all real numbers, $(-\infty, +\infty)$
- **Range:** $y > 0$, i.e., $(0, +\infty)$
- **y-intercept:** At $x = 0$, $y = 2^0 = 1$. The curve always crosses the y-axis at $(0, 1)$.
- **x-intercept:** None — the curve never touches the x-axis.

Table of Values

x	$y = 2^x$	Simplified
-2	$2^{-2} = 1/2^2$	$1/4$
-1	$2^{-1} = 1/2^1$	$1/2$
0	2^0	1
$1/2$	$2^{1/2} = \sqrt{2}$	≈ 1.414
1	2^1	2
2	2^2	4
10	2^{10}	1024

Graph of $y = 2^x$



End Behavior

As $x \to -\infty$

$y \to 0$

The curve approaches — but never touches — the x-axis (horizontal asymptote at $y = 0$)

As $x \to +\infty$

$y \to +\infty$

The curve shoots upward without bound (exponential growth)

Linear vs. Exponential. *Compare $y = 2x + 3$ (a straight line: equal changes in x produce* equal *changes in y) with $y = 2^x$ (a curve: equal changes in x produce* multiplied *changes in y). This difference between* additive *and* multiplicative *growth is one of the most important distinctions in all of mathematics.*

Chapter 8 Practice Worksheet

Exponents, Radicals & Exponential Functions

Name: _____ Date: _____ Score: _____/ 20

Part A Laws of Exponents *(2 pts each)*

Simplify each expression. Leave answers with positive exponents only.

1. $3^2 \cdot 3^5 =$ _____

2. $(5^2)^3 =$ _____

3. $7^0 =$ _____

4. $4^{-3} =$ _____

5. $(2^4)^2 \cdot 2^{-3} =$ _____

Part B Radicals & Rational Exponents *(2 pts each)*

6. Evaluate: $\sqrt[10]{1024} =$ _____

7. Write as a radical: $8^{1/3} =$ _____

8. Write as a rational exponent: $\sqrt[4]{x^3} =$ _____

9. Evaluate: $27^{2/3} =$ _____ *Hint: find the cube root first*

10. True or False: $\sqrt[n]{a} = a^{1/n}$ _____

Part C Exponential Function $y = 2^x$ *(short answer)*

11. Complete the table of values for $y = 2^x$:

x	-3	-1	0	1	3	4
$y = 2^x$						

12. State the domain and range of $y = 2^x$:

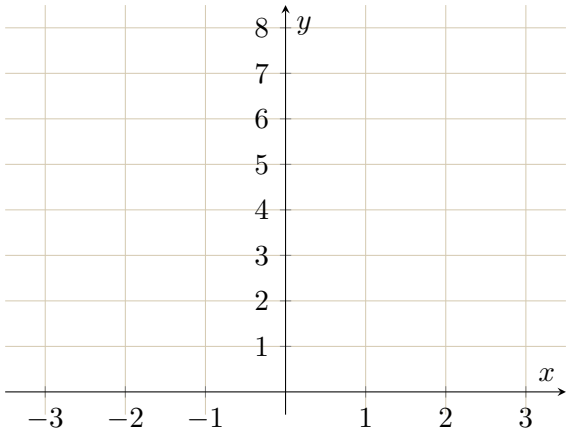
Domain: _____ Range: _____

13. Describe the end behavior of $y = 2^x$ in your own words:

_____

_____

14. Sketch the graph of $y = 2^x$ on the axes below. Label at least three points.



15. *Challenge.* If $y = 2^x$ passes through the point (10, 1024), write this relationship using *radical notation* and using *rational exponent notation*:

Radical: $\sqrt{\quad} =$ _____ Rational exponent: $(1024)^{\quad} =$ _____

Answer Key

Part A — Laws of Exponents

1. $3^7 = 2187$
2. $5^6 = 15{,}625$
3. 1
4. $\dfrac{1}{64}$
5. $2^5 = 32$

Part B — Radicals & Rational Exponents

6. 2
7. $\sqrt[3]{8} = 2$
8. $x^{3/4}$
9. $\left(\sqrt[3]{27}\right)^2 = 3^2 = 9$
10. True

Part C — Exponential Function

11. y values: $\frac{1}{8},\ \frac{1}{2},\ 1,\ 2,\ 8,\ 16$
12. Domain: $(-\infty, +\infty)$; Range: $(0, +\infty)$
13. As $x \to -\infty$, $y \to 0$ (curve approaches but never touches x-axis). As $x \to +\infty$, $y \to +\infty$ (exponential growth).
14. Curve passes through $(-2, \frac{1}{4})$, $(0, 1)$, $(1, 2)$, $(2, 4)$; rises steeply to the right, flattens toward $y = 0$ to the left.
15. Radical: $\sqrt[10]{1024} = 2$; Rational exponent: $1024^{1/10} = 2$